

Exercise 1 - Predict Employee Attrition (Classification)

Kaggle dataset : <https://www.kaggle.com/datasets/pavansubhasht/ibm-hr-analytics-attrition-dataset>

Models to be used : Logistic Regression, SVM, RandomForest, XGBoost

Description:

Step 1: Load & Explore the Data

- Import required libraries (pandas, numpy, matplotlib, seaborn).
- Load the dataset into a DataFrame.
- Display first 5 rows, Shape, Column names and data types
- Generate statistical summary (
- Plot target variable:
 - Histogram of charges
 - Check skewness and outliers
- Visualize distributions and relationships:
 - Pairplot / scatterplots
 - Boxplot for categorical features
- Identify missing values (if any)

Step 2: Data Preprocessing

- Handle missing values (drop / impute — if needed)
- Separate features into. Numerical and Categorical
- Encode categorical features using One-hot encoding
- Apply scaling to numeric features for SVC & Logistic Regression

Step 3: Train - Validation - Test Split

- Perform 80–20 split for validation set and 70–30 for test set.
- Maintain a fixed random_state=42 for reproducibility

Step 4: Train Models

- Measure training accuracy and validation accuracy for all 4 models :
- Logistic Regression, SVM, RandomForest, XGBoost)

Step 5: Hyper-parameter turning with RandomizedSearchCV / GridSearchCV

- Perform cross-validation (cv=5 recommended)

Step 6: Handle Class Imbalance

Step 7: Evaluation

- Confusion matrix, Classification report, AUC-ROC curve, PR curve

Step 8: Feature Importance & Insights

- Extract feature importance from: Random Forest and XGBoost
- Plot feature contributions (bar chart)
- Interpret top contributing features:

Questions to answer:

1. Is the dataset imbalanced? What percentage of employees belong to the “Attrition = Yes” class, and why does this matter in modeling?
2. What are the top 5 features that differ most between Attrition = Yes vs No? What insights do you observe from plots or statistics?
3. Which categorical features required encoding and why? Explain how you encoded them (One-Hot vs Label Encoding) and the reasoning.
4. Why is feature scaling important in this dataset? Which models benefit most from scaling and why?
5. Which model performed better — Logistic Regression or Random Forest? Compare using F1-score and ROC-AUC (not just accuracy).
6. Which model handles class imbalance better and why? Support your answer with precision-recall performance.
7. What did you observe after applying class balancing (e.g., SMOTE or class weights)? Which metric improved and why?
8. What type of errors does your best model still make? Explain using the confusion matrix — which is worse for HR: false positives or false negatives?
9. Which top 3 features contribute the most to attrition according to your model? Explain the business interpretation for each (e.g., “Low job satisfaction increases attrition risk”).
10. What recommendations can you give HR based on model insights? Translate the ML findings into actionable business strategies.

Exercise 2 – Predict Medical Insurance Charges (Regression)

Kaggle dataset: <https://www.kaggle.com/datasets/mirichoi0218/insurance>

Repeat all the steps of Exercise 1 here

Questions to Answer :

1. How are medical costs distributed? Is the target (charges) skewed?
2. Which categorical features needed encoding, and why? Explain your encoding choices (One-Hot recommended).
3. Which features show the strongest correlation with charges? Provide a heatmap and insights.
4. Why is scaling important for some models? Specifically explain for SVR vs tree-based models.
5. Which regression model performed the best and why? Compare using RMSE, MAE, and R^2 .
6. How did hyperparameter tuning improve the model? Show before vs after performance.
7. Does smoking status dramatically increase insurance cost? Provide visual proof (e.g., boxplot).
8. Do older people pay significantly more than younger people? Explain findings based on model interpretation.
9. What are the top 3 most important predictors of cost? Explain them from a business perspective.
10. What policy recommendations can insurers make from your model? Example: Pricing, preventive health measures, customer profiling.
11. Does PCA help on minimising the RMSE ? Explain with evidence.

Exercise 3 – Customer Segmentation (Clustering)

Kaggle dataset: <https://www.kaggle.com/code/utkarshsaxenadn/shop-customer-clustering>

Description :

Step 1: Load & Explore the Data

- Import dataset and preview rows
- Inspect features: check numeric vs categorical
- Visualize distribution of key variables (e.g., Annual Income, Spending Score, Age)

Step 2: Data Preprocessing

- Handle missing values if present
- Standardize/scale numeric features (important for distance-based clustering)
- Possibly select subset of features relevant for segmentation (e.g., Income + Spending Score)

Step 3: Choose & Apply Clustering Algorithm(s)

- Choose at least two methods:
 - K-Means Clustering
 - DB scan
- Fit models and decide on number of clusters (e.g., using elbow method or silhouette method)

Step 4: Evaluate & Validate Clusters

- Use metrics like silhouette score,
- Visualize clusters (scatter plot of two selected features colored by cluster)
- Interpret cluster profiles (avg income, spending, age etc)

Step 5: Feature Importance & Insights

- Describe what distinguishes each cluster: e.g., “High income, low spending”, “Young, high spending”
- Provide actionable business strategy based on segments

Questions to answer:

1. What is the distribution of the main features (Annual Income, Spending Score, Age)?
2. Why is feature scaling important before clustering?
3. Which method did you use to determine the “right” number of clusters (e.g., elbow method, silhouette)? Show the plot.
4. Compare the clusters obtained by K-Means vs Hierarchical clustering. Which seems more meaningful and why?
5. What are the silhouette scores (or other validation metrics) for each method? Which is better?
6. For each cluster, what are the key characteristics (e.g., average income, spending score, age)?
7. How many clusters would you choose for business action, and why?
8. What kinds of business strategies would you recommend for each cluster (e.g., targeted marketing, premium services)?
9. If you added one more feature (e.g., credit card usage), how might that change your segmentation?
10. What are the limitations of your clustering approach and how might you improve it?